

Imperfect Information and Slow Recoveries in the Labor Market

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Abstract

The unemployment rate remains elevated long after recessions, a persistence standard search-and-matching models fail to explain. I show that noise shocks—expectational errors from noisy signals about productivity—constitute a novel channel generating this sluggishness. Using a structural VAR, I find unemployment would have recovered about six quarters earlier absent such shocks. To interpret this evidence, I introduce imperfect information in a search-and-matching model, which successfully replicates the observed recoveries. Persistence arises through two channels: slow learning, in interaction with sticky wages, amplifies the effects of persistent productivity shocks, while noise shocks provide an additional, independent source of sluggishness—highlighting information frictions as an important driver of slow recoveries.

Keywords: Imperfect Information, Labor Market, Business Cycles

JEL Codes: E24, E32, E70

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1 Introduction

One of the long-standing challenges for conventional business cycle models has been to match the persistence of labor market recoveries, especially for the unemployment rate following recessions. In this paper, I document a new stylized fact: professional forecasters systematically overestimate the time that it takes unemployment to recover from a recession, suggesting that ex-ante they expect the recoveries to be even longer than observed. This systematic overestimation suggests the presence of information frictions, where agents misperceive the persistence of aggregate shocks and rely on noisy signals to form expectations. The presence of such frictions offers a mechanism that can slow job creation and search activity, thereby generating a feedback cycle that leads to slower unemployment recoveries. To investigate such a channel, I estimate noise shocks, persistent TFP shocks, and transitory TFP shocks from a tri-variate VAR. I find that the noise shocks play an essential role in explaining the persistence of the labor market. I then incorporate the noise shocks into a model of labor search with imperfect information and confirm that imperfect information is key to the model's success in matching the sluggishness of the unemployment rate during recoveries.

I begin by presenting a set of empirical facts using data on the unemployment rate and forecasts from the Survey of Professional Forecasters (SPF). It takes between 5 and 16 quarters for the unemployment rate in the United States to recover half of its recessionary increase, and more than 20 quarters to return to its pre-recession level. Using forecast data from the SPF, I next document that professional forecasters are even more pessimistic about the recovery than the realized data suggest. There is a consistent wedge between expected and actual unemployment rates across recessions, with forecasts predicting even slower recoveries. This pattern suggests that forecasters, like agents in the economy, may lack perfect information about the aggregate state and instead rely on noisy signals to form expectations about whether changes in fundamentals are persistent or transitory. Such misperceptions can influence key economic decisions, such as hiring and investment, thereby contributing to a sluggish labor market recovery.

To understand these facts, I build upon the identification strategy from Chahrour, Nimark, and Pitschner (2021) and Enders, Kleemann, and Müller (2021) to estimate three shocks from a VAR in utilization-adjusted TFP, real GDP growth, and nowcast errors of professional forecasters: noise shocks, disturbances that arise from changes in expectations without any change in fundamentals; persistent TFP shocks; and transitory TFP shocks. The noise shocks are identified using sign restrictions with the assumption that the noise shocks, by definition, must affect expectations more than the fundamentals. I further

refine this strategy by imposing a zero restriction that rules out any contemporaneous effect of noise shocks on actual TFP. Additionally, to distinguish persistent from transitory TFP shocks, I identify the persistent component as the shock that maximizes the long-run forecast error variance of TFP.

I document two key findings from this exercise. First, labor market variables, including unemployment, vacancies, and job-finding rates, respond significantly to the estimated noise shocks, which account for roughly one-third of their variation at business cycle frequencies. Second, in the absence of noise shocks, unemployment would have recovered 75% of its recessionary rise an average of six quarters earlier. These findings suggest a critical role for the noise shocks in the recovery of the labor market and indicate the presence of information frictions.¹

To study the role of noise shocks in driving labor market persistence, I embed imperfect information (Lorenzoni, 2009) into a model of equilibrium unemployment featuring endogenous search effort, on-the-job search, and sticky wages à la Gertler, Huckfeldt, and Trigari (2020).² The model includes two fundamental productivity shocks, transitory and persistent, but agents cannot distinguish between them. Instead, they observe aggregate productivity and receive a noisy signal about its persistent component, which they use to form beliefs about future productivity. Noise shocks add complexity by making it harder for agents to discern whether observed changes reflect true persistent shocks or simply noise in the signal. The calibrated model successfully accounts for the sluggish labor market recovery: relative to a full-information counterpart, the imperfect-information model extends the recovery of unemployment by about five quarters (about 30%).

There are two channels for the success of the model. First, learning under imperfect information generates persistence endogenously: it takes time for agents to learn whether a shock is persistent or not, leading to an initial under-reaction. Wage stickiness further slows adjustment, as infrequent renegotiation makes agents hesitant to revise wages without fully knowing the shock's persistence. This interaction between imperfect information and wage rigidity is novel and central to the persistence mechanism. Second, noise shocks themselves may prolong recoveries, as agents mistake them for actual productivity shocks and respond to them accordingly, even though the fundamentals are unchanged. Together,

¹If firms and workers had perfect information about the shock being noise, noise shocks would not exist and it would not be optimal for agents to respond to it. However, initially agents misperceive the noise shock as an actual negative productivity shock and hence firms decrease their hiring. As a result, there are fewer job opportunities for workers and the job-finding rate decreases. This contributes to an increase in unemployment.

²Pissarides (2009) critiques models that rely on sticky wages for new hires. Several papers provide responses to this critique: Gertler, Huckfeldt, and Trigari (2020) show that staggered Nash bargaining can reconcile wage stickiness with data on unemployment fluctuations; Grigsby, Hurst, and Yildirmaz (2021) and Hazell and Taska (2025) present empirical evidence of wage rigidity in administrative and vacancy-level data. An alternative view is offered by Kudlyak (2014), who emphasizes the cyclicity of the user cost of labor rather than wages directly.

these mechanisms successfully account for the persistence in the unemployment dynamics following downturns.

Further, the model can replicate the systematic overestimation of medium-run unemployment by professional forecasters observed in the data. This overshooting arises endogenously from agents' misperceptions driven by noise shocks and cannot be replicated in models with only structural productivity shocks, which, by definition, imply that actual outcomes exceed expectations and thus tend to generate forecast under-reaction, a well-known limitation of standard macroeconomic models. This ability to match both the persistence in unemployment and the forecast errors constitutes not only a success for the model but is also a key empirical validation of the imperfect information channel.

This paper builds on the literature documenting the persistence of unemployment following recessions and the difficulty conventional business cycle models face in explaining this fact, as documented by Cole and Rogerson (1999), and more recently by Hall and Kudlyak (2022) and Ferraro (2023). Slow labor market recoveries have been a consistent feature of postwar recessions, yet there remains no unified consensus on their underlying drivers. Prominent explanations include job polarization (Jaimovich and Siu, 2020), organizational restructuring following long expansions (Berger et al., 2012; Koenig and Rogerson, 2005), changes in the persistence of business cycles (Bachmann, 2012; Panovska, 2017), the rising importance of technology shocks since the mid-1980s (Barnichon, 2010b), extended unemployment insurance (Mitman and Rabinovich, 2024), and the convergence of female employment (Fukui, Nakamura, and Steinsson, 2023). These explanations focus on evolving fundamentals. In this paper, I instead document an important role for noise shocks that arise when agents face imperfect information about changes in fundamentals. This mechanism complements the existing literature by showing how informational frictions, when embedded in a search and matching model, can generate sluggish unemployment recoveries without requiring large changes in fundamentals.

A separate strand of the literature emphasizes asymmetric labor market dynamics, where recessions trigger sharp rises in unemployment but recoveries are slow. This asymmetry has been modeled through endogenous job separation (Andolfatto, 1997) and worker heterogeneity in productivity (Ferraro, 2018). These models generate slow recoveries by amplifying shocks through non-linearities in the labor market. My paper instead documents new evidence that the underlying driving forces differ systematically between recessions and expansions, pointing to a complementary mechanism. In particular, I show that professional forecasters systematically overestimate unemployment during recoveries—a stylized fact that conventional models cannot replicate.

This paper therefore connects to the long-standing research agenda on information

frictions in macroeconomics, dating back to Lucas (1972, 1975) and developed further in Angeletos and La'O (2010, 2013), Blanchard, L'Huillier, and Lorenzoni (2013) and Angeletos, Collard, and Dellas (2020). These theories demonstrate how noisy signals about fundamentals can distort beliefs and propagate shocks. My contribution is to bring these information frictions into a quantitative search-and-matching model of the labor market and evaluate their implications for unemployment persistence. In parallel, an empirical literature has developed methods for identifying noise shocks from macroeconomic data, most recently Chahrour, Nimark, and Pitschner (2021) and Enders, Kleemann, and Müller (2021). I advance this empirical agenda by refining their identification strategy: I impose additional restrictions and identify persistent and transitory productivity shocks separately, which allows me to quantify the distinct role of noise shocks in labor market recoveries. Taken together, these contributions connect the information-frictions and labor-market literatures and provide a unified empirical-quantitative account of sluggish recoveries.

Within the labor market literature, Venkateswaran (2014) shows that firms' inability to disentangle aggregate from idiosyncratic shocks can generate volatility far larger than in standard search models. In that framework, firms misattribute part of aggregate shocks to idiosyncratic factors and, because they respond more strongly to the latter, hiring fluctuations are amplified, helping resolve the Shimer (2005) volatility puzzle. My paper complements this by focusing on a different friction: imperfect information about the persistence of aggregate shocks. Whereas Venkateswaran (2014) emphasizes volatility driven by misperceptions of the source of shocks, I show that noise shocks tied to misperceptions of persistence generate sluggish labor market recoveries and systematic forecast errors.

More recent work by Faccini and Melosi (2022) and Morales-Jiménez (2022) quantitatively assesses the importance of information frictions in labor market dynamics. Further, Kozłowski, Veldkamp, and Venkateswaran (2020) and D'Agostino, Mendicino, and Puglisi (2022) show that imperfect information about the distribution of shocks can produce persistent effects from transitory disturbances, particularly during the Great Recession. My analysis complements these papers by combining new empirical evidence—systematic forecast errors by professional forecasters—with a structural model that demonstrates how noise shocks and imperfect information jointly account for sluggish labor market recoveries across recessions. This paper also relates to the growing literature incorporating subjective beliefs into business cycle models (Bhandari, Borovička, and Ho, 2025). In particular, Menzio (2023) shows that stubborn beliefs can generate persistent effects in search equilibrium; my analysis complements this by identifying the empirical source of

such belief distortions and quantifying their role in unemployment recoveries.

A complementary line of research emphasizes how unemployment risk feeds back into aggregate demand through household consumption. When workers perceive a higher risk of job loss, they cut spending and increase precautionary savings, which can amplify downturns and slow recoveries even in the absence of large movements in productivity (Kekre, 2023; McKay and Reis, 2016; Ravn and Sterk, 2021). In this paper I largely abstract from these aggregate demand channels and instead hold the demand side fixed, focusing on how imperfect information about productivity interacts with wage rigidity and hiring frictions to shape unemployment dynamics. The mechanisms are therefore complementary: demand-driven amplification operates through consumption and monetary policy, while my mechanism operates through beliefs about fundamentals and the labor market's adjustment to those beliefs.

The rest of the paper is organized as follows. Section 2 presents the empirical evidence and identifies noise shocks using a structural VAR, and documents their effects on labor market dynamics. Section 3 develops a general-equilibrium search-and-matching model with imperfect information. Section 4 describes the calibration and estimation strategy. Section 5 discusses the quantitative results and counterfactual exercises. Finally, Section 6 summarizes the main findings and discusses implications and directions for future work.

2 Unemployment Recoveries and Noise Shocks

This section documents two empirical facts that motivate the mechanism. First, unemployment recoveries in the United States are slow: between 1968 and 2019, returning to the pre-recession trough takes almost five years on average. Second, professional forecasters often expect recoveries to be even slower than they are. I then identify noise shocks using GDP nowcast errors and show that these shocks have persistent effects on unemployment, vacancies, and job-finding rates; absent noise shocks, unemployment would have recovered 75% of its recessionary rise an average of six quarters earlier.

Unemployment Dynamics During Recoveries U.S. labor market recoveries are typically slow, with unemployment remaining elevated well after job destruction subsides. To summarize recovery dynamics consistently across episodes, I follow Heise, Karahan, and Şahin, 2022 and compute the share of the recessionary increase in unemployment that has been reversed during the subsequent expansion:

$$(1) \quad URecovery_t = \frac{u_{peak} - u_t}{u_{peak} - u_{trough}}.$$

Here, u_{peak} is the recession peak and u_{trough} is the pre-recession trough in the unemployment rate.

Figure 1 and Table 1 summarize these recovery patterns across post-1968 recessions. Recoveries are slow and have become slower over time: the unemployment rate takes about 11 quarters on average to reverse half of its recessionary increase and about 23 quarters to return fully. Before 2000, the 50% threshold is reached in about 9 quarters; after 2000, it takes about 14 quarters.

Figure 1: Unemployment Recovery Across Recessions

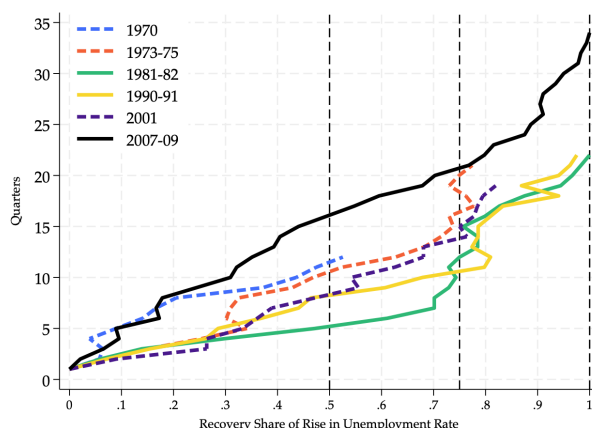


Table 1: Unemployment Recovery Across Recessions

	(1)	(2)	(3)	(4)	(5)
Recessions	25%	50%	75%	100%	NBER
2007-09	10	17	22	33	25
2001	5	11	18	NA	40
1990-91	5	9	14	22	31
1981-82	4	5	9	14	4
1973-75	4	11	20	NA	12
1969-70	9	12	NA	NA	36
Average:					
Total	6	11	17	23	30
Pre 2000	6	9	13	18	21
Post 2000	8	14	20	32	33

Note: Entries report the number of quarters required to recover 25%, 50%, 75%, and 100% of the recessionary increase in unemployment. The 1980 recession is excluded because it is immediately followed by the 1981–82 recession.

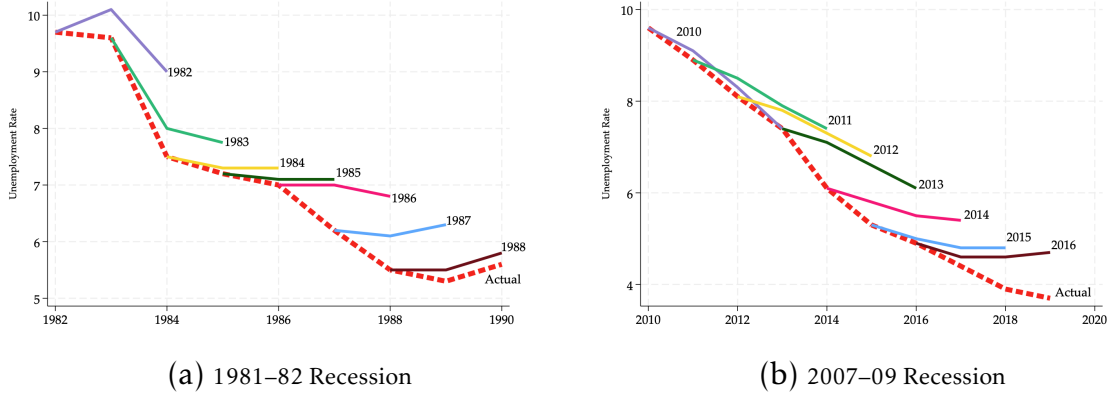
Forecast Errors and Misperception about the Evolution of the Unemployment Rate.

While unemployment remains elevated during recoveries, forecasters often expect even slower labor-market adjustment. Figure 2 shows this pattern for the 1981–82 recession using Livingston Survey projections and for the Great Recession using SPF long-run projections.³ Online Appendix Figure A5 shows the broader SPF one-year-ahead series. These forecast patterns suggest a persistent wedge between expected and realized unemployment around recoveries.

One potential explanation is imperfect information about whether changes in aggregate fundamentals are persistent or transitory (Edge, Laubach, and Williams, 2007). I study this mechanism by identifying an aggregate noise shock from GDP *nowcast errors*, following the logic that deviations between realized outcomes and real-time expectations contain information about misperceptions (Enders, Kleemann, and Müller, 2021), and

³The SPF reports long-run unemployment projections only from 2009 onward, so I use the Livingston Survey for earlier long-horizon expectations.

Figure 2: Unemployment Rate: Projections and Actual



Note: In Panel (a), the various colored lines represent the median 1-year and 2-year ahead projections of the unemployment rate from the Livingston Survey. The solid red line is the actual unemployment rate during the 1981–82 recession. In Panel (b) the various colored lines represent the median long-run (1-year, 2-year, and 3-year ahead) projections of the unemployment rate from the Survey of Professional Forecasts during the Great Recession. The dashed red line is the actual unemployment rate.

then estimating how labor-market outcomes respond.

Nowcast Errors The nowcasts are SPF median expectations about current-quarter GDP growth. I define nowcast errors as realized growth minus the contemporaneous median forecast:

$$(2) \quad nce_t = \Delta y_t - E_t^{median}(\Delta y_t)$$

where Δy_t is realized real GDP growth. Because SPF respondents report current-quarter expectations before current output is observed, nce_t is available only ex post and gives the econometrician information not available to agents in real time. These errors identify noise shocks under the restriction that noise moves expected GDP more than realized GDP on impact.

An alternative explanation for the forecast patterns is asymmetric loss functions: forecasters may prefer to overpredict unemployment to avoid costly under-predictions. If this were the primary driver, however, we would expect systematic overprediction during expansions as well. Online Appendix Figure A5a (panel (a) of Figure A5) shows that overprediction is concentrated around recoveries and does not persist systematically during expansions.

2.1 Identification of Noise Shocks

I identify a persistent TFP shock, a transitory TFP shock, and a noise shock in two steps. First, I estimate a tri-variate SVAR in TFP, real GDP growth, and GDP nowcast errors. Second, I use the identified shocks in local projections to estimate their effects on unemployment, vacancies, worker flows, and wages.

Empirical Specification. The aggregate productivity process has persistent and transitory components that agents cannot separately observe. Noise shocks are non-fundamental movements in signals about productivity. I estimate the VAR

$$(3) \quad A_0 \mathbf{Y}_t = a + \sum_{j=1}^p A_j \mathbf{Y}_{t-j} + \mathbf{e}_t$$

where $\mathbf{Y}_t \equiv [TFP_t, GDP_t, NCE_t]$ includes utilization-adjusted TFP from Fernald, 2014, real GDP growth, and GDP nowcast errors over 1968Q4–2019Q4.⁴ Nowcast errors are observed by the econometrician only ex post, which provides the informational advantage needed for invertibility (Blanchard, L’Huillier, and Lorenzoni, 2013). Online Appendix Section A.4 provides additional implementation details and diagnostic results.

Identification Assumptions Aggregate noise shocks are identified using zero and sign restrictions, while the persistent TFP shock is identified using a max-share restriction.

1. I impose the following impact restrictions to identify the noise shock.
 - (a) Noise shocks have zero contemporaneous impact on TFP.
 - (b) Persistent and transitory TFP shocks move TFP and GDP growth in the same direction on impact, while their effect on nowcast errors is unrestricted.
 - (c) Noise shocks move GDP growth and nowcast errors in opposite directions, as in Enders, Kleemann, and Müller, 2021 and Chahrour, Nimark, and Pitschner, 2021.

These identifying restrictions hold across a broad class of models with information structures consistent with Lorenzoni, 2009, Blanchard, L’Huillier, and Lorenzoni, 2013, and Angeletos and La’O, 2010. Let ϵ_t be the persistent shock, η_t be the transitory shock and ν_t be the noise shock. Thus, the restrictions on the impact matrix can be demonstrated by the following:

⁴The VAR excludes labor-market outcomes, so the identified shocks are not mechanically driven by unemployment or vacancy fluctuations.

$$(4) \quad \begin{bmatrix} TFP_t \\ GDP_t \\ NCE_t \end{bmatrix} = \sum_{j=1}^p B_j \begin{bmatrix} TFP_{t-j} \\ GDP_{t-j} \\ NCE_{t-j} \end{bmatrix} + \begin{bmatrix} + & + & 0 \\ + & + & + \\ * & * & - \end{bmatrix} \begin{bmatrix} \epsilon_t \\ \eta_t \\ \nu_t \end{bmatrix}$$

2. The sign restrictions identify the noise shock but do not distinguish persistent from transitory productivity shocks. I therefore extract the persistent shock as the innovation that maximizes the forecast-error-variance share of TFP at a long but finite horizon, following the max-share logic in Uhlig et al., 2004, Francis et al., 2014, and Kurmann and Sims, 2021.

The identifying rotation $\mathbf{Q}$ solves

$$(5) \quad \mathbf{Q}^* = \arg \max_{\mathbf{Q} \in \Theta(3)} \mathbf{q}_1' \Gamma_H^1(\phi) \mathbf{q}_1$$

subject to

$$(6) \quad \mathbf{e}_1' \mathbf{P} \mathbf{q}_3 = 0$$

$$(7) \quad \mathbf{S}_j(\phi) \mathbf{q}_j \geq \mathbf{0}, \text{ for } j \in \mathcal{I}_S$$

$$(8) \quad \mathbf{Q}' \mathbf{Q} = \mathbf{I}_3$$

where $\mathbf{e}_1 = (1, 0, 0)'$ selects TFP on impact and Γ_H^1 is the TFP forecast-error-variance matrix. The transitory TFP shock is the remaining productivity shock consistent with these restrictions and orthogonality.

I set $H = 40$ quarters and use four lags as suggested by the Akaike Information Criterion. To extend the shock series after 2012, I use the maximum available horizon in later years; Online Appendix Figure A9 shows that persistent shocks still explain the largest TFP variance share at shorter horizons. I solve the rotation problem using the algorithm in Carriero and Volpicella, 2025.

Online Appendix Figure A8 reports the resulting SVAR impulse responses. Nowcast errors respond persistently to fundamental shocks, consistent with gradual learning about shock type, while noise shocks move forecast errors and GDP without moving TFP on impact.

2.2 Effect of Noise Shocks on Labor Market Dynamics

Key labor market variables exhibit significantly persistent impulse responses to the identified noise shocks at the business cycle frequency (8-10 quarters). A historical decomposition shows that noise shocks have an increasingly important role to play in the evolution of unemployment, vacancies and job-finding rates over the business cycle, which is a motivation to introduce imperfect information in a search and matching model.

Local Projections Once the shocks are extracted from the VAR, I estimate responses of labor-market variables using smooth local projections (SLP) following Barnichon and Brownlees (2019). The SLP estimator smooths horizon-specific local-projection coefficients with B-splines, reducing sampling noise while preserving the flexibility of local projections; details are in Online Appendix Section A.5. All empirical IRFs used in the model calibration are estimated with this same specification.

For each identified structural shock series e_t^j , the Jordà (2005) local projections are given by

$$(9) \quad y_{t+h} = \alpha_h^j + \beta_h^j e_t^j + \sum_{p=1}^P \delta_p^j \omega_{t-p} + \mu_{h,t+h}^j$$

where ω_{t-p}^j is the vector of lagged controls, including lags of y_t and the identified shock e_t^j .

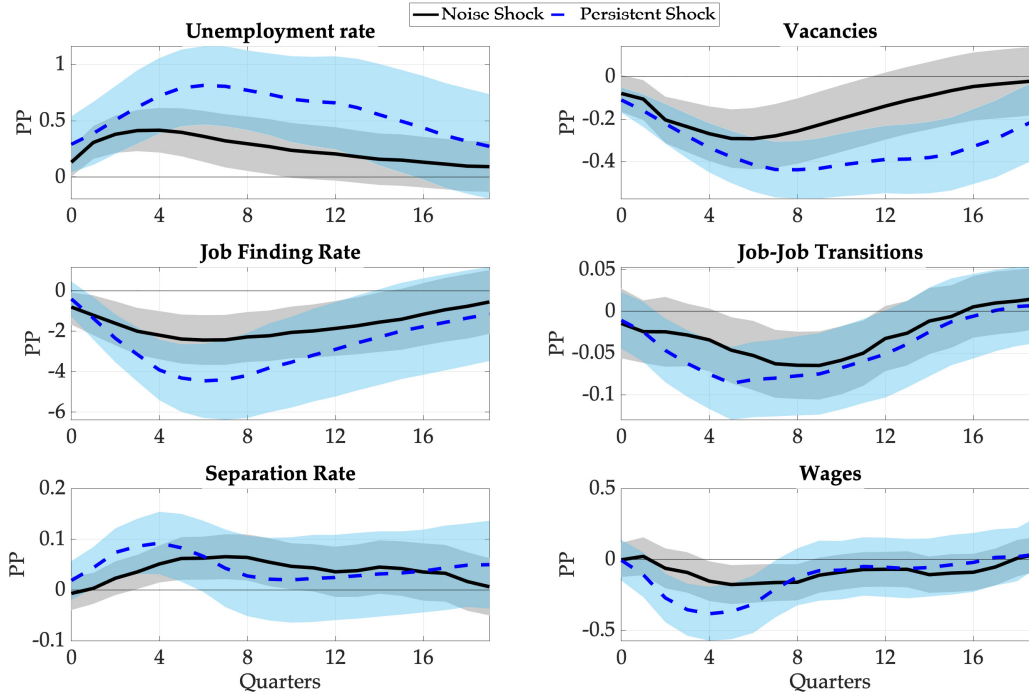
Here, y_t denotes aggregate labor market outcomes such as the unemployment rate, vacancies, the rate of outflow from unemployment (UE), job-to-job transition rates (EE), the hiring rate, and wage growth. e_t^j denotes the identified shock series for shock j , while μ_t^j is the residual error for each regression. All labor market data are from the Current Population Survey, with vacancies and the hiring rate from JOLTS.⁵ Figure 3 shows the impulse responses of the labor market variables to standardized one-standard-deviation negative noise and persistent TFP shocks.⁶

Noise shocks have significant effects on unemployment, vacancies, UE , EE , and the hiring rate for up to 10 quarters. A one-standard-deviation noise shock raises unemployment by about 0.4 percentage points, lowers vacancies and job-finding rates, and generates a delayed, weak response of wages. These responses are consistent with gradual information updating: agents initially treat part of the noise innovation as a deterioration in fundamentals, reducing hiring and search until beliefs adjust. Section 5 provides structural discipline for this interpretation.

⁵The results are consistent with using an Autoregressive Distributed Lag (ADL) specification for local projections.

⁶Response to the transitory TFP shocks is documented in Online Appendix Figure A12.

Figure 3: Impulse Response to Persistent TFP Shocks



Note: This figure shows the smoothed cumulative impulse response functions for key labor market variables to a persistent TFP shock (in blue), estimated using equation 9, where u_j is the persistent TFP shock identified using the SVAR described by the optimization problem in equation 5. The 95% confidence interval is shaded in blue as well. It is superimposed on the IRFs from the noise shocks. The sample period is 1968Q4–2019Q4. Data for the labor market outcomes are from CPS, vacancies from Barnichon, 2010a and wages from BEA’s average hourly earnings series.

These results motivate introducing imperfect information into a general-equilibrium search-and-matching model: under complete information, firms and workers would not respond persistently to shocks that do not change fundamentals.

Forecast Error Variance Decomposition I compute local-projection forecast error variance decompositions using the coefficient-of-determination estimator in Gorodnichenko and Lee, 2020. Table 2 shows that at horizons of 0–8 quarters, noise shocks explain 34% of unemployment variation, 37% of vacancy variation, 35% of UE variation, 27% of EE variation, and 14% of wage-growth variation.

At 8–16 quarters, persistent shocks account for most of the variance, but noise shocks remain nontrivial for unemployment, vacancies, UE , and EE transitions. Thus, noise shocks are especially important at business-cycle horizons, while persistent fundamentals dominate at longer horizons.

Table 2: Forecast Error Variance Decomposition: Shorter Run Horizon

	Short Run			Medium Run		
	Horizon: 0-8 quarters			Horizon: 9-16 quarters		
	Persistent	Transitory	Noise	Persistent	Transitory	Noise
Unemployment	0.43	0.23	0.34	0.63	0.21	0.16
Vacancies	0.42	0.21	0.37	0.61	0.20	0.19
UE	0.38	0.27	0.35	0.63	0.20	0.17
EE	0.42	0.31	0.27	0.65	0.16	0.19
Wage Growth	0.61	0.25	0.14	0.92	0.05	0.03

Note: This table reports the average forecast error variance decomposition for U , V , $E-E$, $U-E$ and ΔW over a short-run (0–8 quarters) and a medium-run (8–16 quarters) horizon. Each row adds to 1. Noise shocks explain a significant variation in the labor market at a short-run horizon. The sample period is 1968Q4–2019Q4. Data for the labor market outcomes are from CPS, vacancies from Barnichon, 2010a and wages from BEA’s average hourly earnings series.

Historical Contribution of Noise Shocks. To quantify the role of noise shocks over the business cycle, I decompose labor-market outcomes into the cumulative contributions of the three identified shocks:

$$(10) \quad y_t - \bar{\Psi}_t = \sum_j \sum_{h=0}^{t-t_0} \beta_h \cdot \mu_{j,t-h}$$

where $\bar{\Psi}_t$ is the deterministic component and y_t denotes unemployment, vacancies, job-finding, or wages.

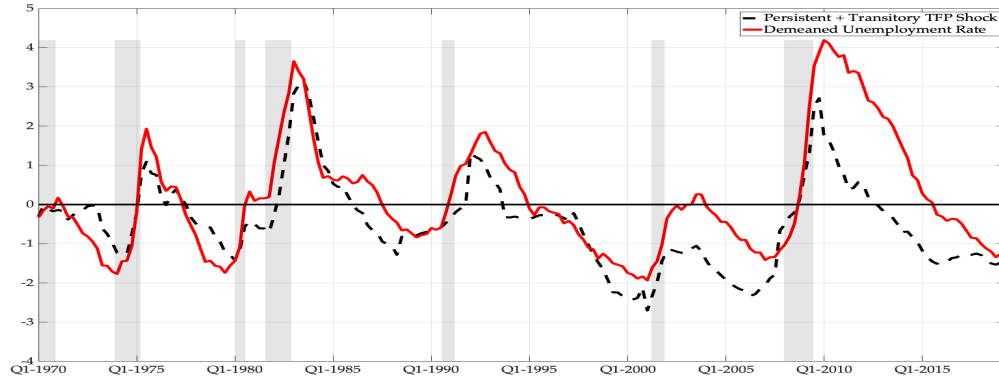
Two key facts emerge from the historical decomposition: first, that the productivity shocks alone fail to account for the persistence of the unemployment rate after 1985, and second, that noise shocks have been playing an increasingly important role since the 1990s. Figure 4a plots the deviation of unemployment rate from its predicted path due to the persistent and transitory productivity shocks alone. Thus, the remaining movement is explained by the noise shocks.

Across the 1990–91, 2001, and 2007–09 recessions, productivity shocks alone predict faster recoveries and, in the Great Recession, a smaller unemployment peak. Outflows from unemployment and vacancies show the same pattern: during downturns, noise shocks amplify the decline in job finding and vacancy posting. These decompositions imply that firms and workers perceived recessions as more persistent than fundamentals warranted, contributing to elevated and persistent unemployment.

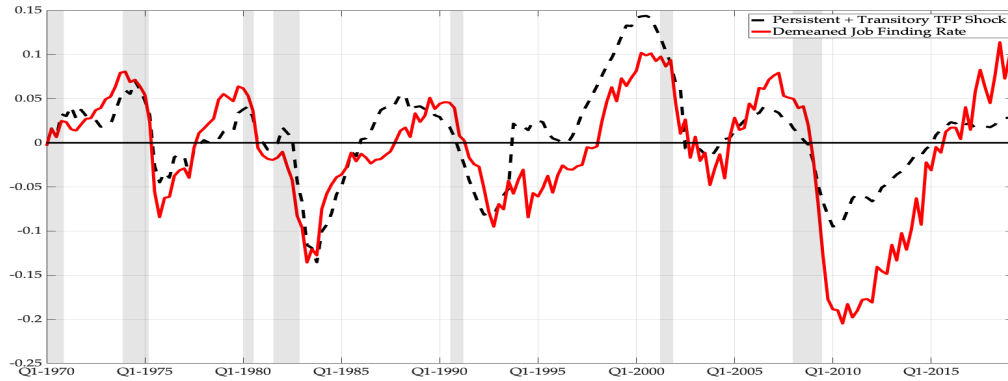
Using the same 75% recovery metric as in Equation 1, Online Appendix Table A2 shows that noise shocks account for about 35% of the Great Recession recovery duration

Figure 4: Historical Contribution of Persistent and Transitory TFP Shocks

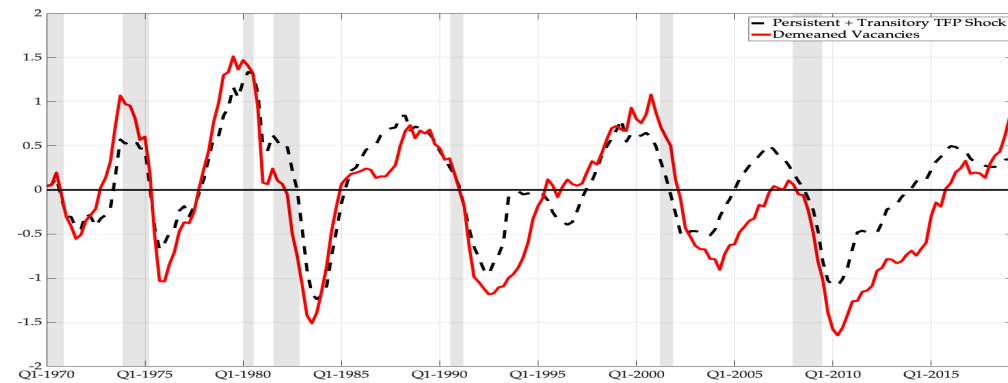
(a) Unemployment Rate



(b) Outflow from Unemployment



(c) Vacancies



Note: This figure shows the historical decomposition of unemployment rate, vacancy postings and outflow rate from unemployment following equation 10. The dashed black line is the cumulative contribution of the identified persistent and transitory TFP shocks to the movements in the demeaned vacancy postings (solid red line). The remaining movement is explained by the noise shocks, which contribute significantly to the vacancy postings during the recessions in 1990–91, 2001 and 2007–09.

and 32% on average across recessions. Their role is larger after 2000: noise shocks explain 19% of recovery duration before 2000 and 36% afterward.

The empirical results indicate that noise shocks play a significant role in explaining the dynamics of the labor market over the business cycle and specifically the sluggish recovery from recessions. Now, to understand the mechanism through which noise shocks affect the persistence of the labor market, I introduce imperfect information in a general equilibrium model of search and matching in the following section.

2.3 Robustness

I next summarize robustness to omitted shocks, alternative interpretations of forecast errors, and the identification strategy; further robustness checks appear in Online Appendix Section A.

Inclusion of Unrestricted Shocks To address the concern that omitted monetary, financial, or other shocks are absorbed by the three-shock system, I estimate a four-variable VAR that adds unemployment and allows a fourth unrestricted shock. Online Appendix Figure A18 shows that noise shocks continue to explain a significant part of unemployment fluctuations in the 1990–91, 2001, and 2007–09 recessions, while the unrestricted shock matters more for the 1970s and early-1980s recessions. I also control for uncertainty shocks following Bloom (2009); the resulting noise-shock IRFs remain within the baseline confidence bands (Online Appendix Section A.7).

Natural-rate misperceptions. A plausible concern is that part of the unemployment forecast error reflects slow-moving misperceptions about the natural rate, rather than only misperceived cyclical persistence. Online Appendix Figure A4 compares standard unemployment forecast errors with a gap-adjusted measure that nets out disagreement between SPF long-run unemployment expectations and the CBO natural rate. This adjustment reduces average forecast errors by about one-third but does not eliminate them, leaving a sizable residual consistent with misperceived cyclical persistence.

Distinguishing Noise from News Shocks. A second concern is that the identified shocks may capture news about future productivity rather than pure noise. The long-run TFP identification and the estimated TFP impulse responses support the noise interpretation: the identified noise shock moves forecast errors, GDP, and labor-market variables, but has zero impact on TFP and does not generate systematic medium-run TFP movements; further discussion appears in Online Appendix Section A.4.

3 A Search and Matching Model with Information Frictions

Standard models of search-and-matching fail to fully explain the volatility of unemployment and vacancies as well as the slow recovery of the labor market from recessions Cole and Rogerson, 1999.⁷ Motivated by the empirical evidence that imperfect information plays an important role in explaining these dynamics, this section introduces an imperfect information structure to a search and matching model.

The model is based on a real business cycle model with search and matching in the labor market as in Merz, 1995 and Andolfatto, 1996 and follows the extensions by Gertler, Huckfeldt, and Trigari, 2020 who introduce staggered wage contracting and allow for on-the-job search with variable intensity. The primary reason for introducing staggered wage contracting in the context of this paper is that wage rigidity amplifies the role of imperfect information, as will become clear in the subsequent subsections (Chahrour and Jurado, 2018; Morales-Jiménez, 2022). The reason for introducing endogenous search effort as well as on-the-job search is to capture the response coming from workers when faced with information frictions. Job-to-job transitions capture not only the cyclical wage gains, but also crowd out unemployed workers searching for a job, thus capturing an important moment of the labor market. In the following subsections I describe the environment for the model and discuss the problems of firms and households.

Environment There is a continuum of firms and workers, each of measure unity. Firms that post vacancies and workers looking for jobs meet randomly. The aggregate productivity in the economy is given by Z_t . Idiosyncratic match quality is revealed once a worker and a firm meet. Match quality within a firm is either good (g) with probability ξ or bad (b) with probability $1 - \xi$. The productivity of a bad match is a fraction ϕ of the productivity of a good match, where $\phi \in (0, 1)$. Let g_t and b_t denote, respectively, the firm-level stocks of good and bad matches. The firms' effective labor force is

$$(11) \quad l_t = g_t + \phi b_t.$$

⁷As Cole and Rogerson, 1999 note, the DMP model can account for business cycle facts only if the average duration of non-employment spells is nine months or longer, which is quite high relative to that observed in the data.

The total number of unemployed workers is given by:

$$\begin{aligned}\bar{u}_t &= 1 - \bar{g}_t - \bar{b}_t, \\ \bar{g}_t &= \int_i g_t di, \\ \bar{b}_t &= \int_i b_t di,\end{aligned}$$

where $\bar{g}_t$ and $\bar{b}_t$ are the total number of workers in good and bad matches respectively across all firms, indexed by i .

Workers search for jobs when they are unemployed with endogenous search intensity s_{ut} . Employed workers in a bad match also search on the job so that they can move up the ladder and match with a good job. They search with endogenous search intensity s_{bt} . Workers searching on the job only transition to good jobs. If they are matched with another bad job, they stay in their current bad jobs and hence lateral movements to other bad jobs are eliminated.⁸ Search is costly and the cost of searching is characterized by

$$c(s_{jt}) = \mu(s_{jt})^{1+\frac{1}{\omega}},$$

where s_{jt} is the search intensity of unemployed workers ($j = u$) and employed workers in bad matches ($j = b$). There are two ways a match can be dissolved. First, firms and workers may receive an exogenous separation shock with probability $1 - \sigma$. Workers who receive the separation shock become unemployed at the beginning of the next period. Second, if the match is not destroyed, a worker in a bad match searches on the job. If she finds another job and accepts it, the worker moves to the new firm within the period and the match with the current employer is dissolved. The total efficiency units of search are therefore given by the search-intensity-weighted sum of searchers

$$\bar{s}_t = s_{ut}\bar{u}_t + \sigma s_{bt}\bar{b}_t.$$

The aggregate number of matches is thus a function of the efficiency-weighted number of searchers $\bar{s}_t$ and the number of vacancies $\bar{v}_t$:

$$\bar{m}_t = \Psi \bar{s}_t^\alpha \bar{v}_t^{1-\alpha},$$

where α is the elasticity of matches to units of search and Ψ is the matching efficiency.

⁸As Gertler, Huckfeldt, and Trigari, 2020 explain, the expected gain from a lateral move is quantitatively trivial and can be ruled out with a small moving cost.

The probability that a unit of search leads to a match is given by

$$p_t = \frac{\bar{m}_t}{\bar{s}_t}.$$

It follows that the probability that the match is good (p_t^g) or bad (p_t^b) is given, respectively, as follows:

$$\begin{aligned} p_t^g &= \xi p_t, \\ p_t^b &= (1 - \xi) p_t. \end{aligned}$$

For a firm, the probability that a vacancy will lead to a match is:

$$q_t^m = \frac{\bar{m}_t}{\bar{v}_t}.$$

Now, not all matches will lead to hires since I assume that workers in bad matches accept only good jobs. Thus, the probability that a vacancy leads to a good-quality hire (q_t^g) or to a bad-quality hire (q_t^b) is given by

$$\begin{aligned} q_t^g &= \xi q_t^m, \\ q_t^b &= (1 - \xi) \left(1 - \frac{\sigma s_{bt} \bar{b}_t}{\bar{s}_t} \right) q_t^m. \end{aligned}$$

Since all workers accept good matches, q_t^g is simply the product of the probability of a match being good conditional on a match and the probability of a match. However, since workers in bad matches do not make lateral movements, the fraction of searchers who search on-the-job from bad matches, $\frac{\sigma s_{bt} \bar{b}_t}{\bar{s}_t}$, is netted out to calculate q_t^b . Thus, the expected number of workers in efficiency units of labor that a firm can expect to hire from posting a vacancy is:

$$q_t = q_t^g + \phi q_t^b.$$

Thus, the total number of new hires (in efficiency units) is $q_t v_t$ and the hiring rate χ_t is the ratio of new hires to the existing stock l_t , given by:

$$\chi_t = \frac{q_t v_t}{l_t}.$$

We can now define some laws of motion for the good and bad matches, respectively:

$$(12) \quad \bar{g}_{t+1} = \sigma \bar{g}_t + \xi p_t \bar{s}_t,$$

$$(13) \quad \bar{b}_{t+1} = \sigma(1 - s_{bt} \xi p_t) \bar{b}_t + (1 - \xi) p_t s_{ut} \bar{u}_t.$$

Total good matches next period are the sum of surviving good matches in the current period and an inflow of searchers into good matches, which depends on their probability of finding a good match. Similarly, the total number of bad matches next period is the sum of two terms. The first term represents the number of workers in bad matches who are unable to find a good match and thus remain in the bad match. The second term is the number of unemployed workers who search with intensity s_{ut} , find bad matches, and move into them. The current values of l_t , g_t and b_t are predetermined state variables.

The intra-period timing protocol that the firm's and household's decision problems are based upon is as follows. At the beginning of period t , given the information set $\mathcal{I}^{t-1}$, firms choose vacancies and capital, and households choose consumption, savings and search intensities. During period t , aggregate and idiosyncratic productivity shocks are realized and production takes place given these predetermined choices. Next, match-specific separation shocks are realized. Finally, surviving workers search and new matches are formed, which become productive in period $t + 1$.

Information Structure Agents do not observe the persistent component of productivity x_t directly when they choose period- t controls. Instead, they begin period t with information accumulated through the end of period $t - 1$ and use that information to forecast current aggregate conditions. They only observe the current realization of productivity and the current public signal at the end of period t , after choices for period t have already been made.

Information sets and timing. Let

$$\mathcal{I}^{t-1} \equiv \mathcal{I}_{t-1}^{\text{end}}$$

denote the information set available at the beginning of period t . This set contains the history of aggregate productivity and public signals up to period $t - 1$, together with past endogenous variables:

$$\mathcal{I}^{t-1} = \{z_\tau, \hat{s}_\tau, \text{endogenous variables at } \tau \leq t - 1\}.$$

Expectations used for period- t decisions are therefore

$$\mathbb{E}_t[\cdot] \equiv \mathbb{E}[\cdot \mid \mathcal{I}^{t-1}].$$

In particular, firms and households choose k_t , vacancies, savings, and search effort at the beginning of period t as functions of $\mathcal{I}^{t-1}$. Hence they do not condition these choices on the current realizations $(z_t, \hat{s}_t)$; instead, they condition on their forecast of period- t conditions formed using information through $t - 1$.

Productivity and signal process. Let $z_t = \log Z_t$. Aggregate productivity has a persistent component x_t and a transitory component η_t :

$$(14) \quad z_t = x_t + \eta_t \quad ; \quad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2).$$

The persistent component follows

$$(15) \quad x_t = \rho x_{t-1} + \epsilon_t \quad ; \quad \epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2).$$

Each period, agents receive a public signal about x_t :

$$(16) \quad \hat{s}_t = x_t + a_t,$$

$$(17) \quad a_t = \rho_a a_{t-1} + v_t \quad ; \quad v_t \sim \mathcal{N}(0, \sigma_v^2).$$

The shocks η_t , ϵ_t , and v_t are mutually independent. The shock ϵ_t changes fundamentals and therefore affects both realized productivity and beliefs. The shock η_t changes realized productivity only temporarily, but it can still affect beliefs because agents do not perfectly distinguish temporary from persistent movements. The shock v_t changes the signal without changing productivity, so it is a pure belief shock. Because the noise component a_t is persistent, agents must disentangle whether a persistent movement in observables reflects true persistent productivity or persistent noise in the signal.

Belief updating. At the beginning of period t , before observing current outcomes, agents hold the prior

$$x_{t|t-1} \equiv \mathbb{E}[x_t \mid \mathcal{I}^{t-1}].$$

After period- t outcomes are realized, agents observe $(z_t, \hat{s}_t)$ and the information set becomes

$$\mathcal{I}_t^{\text{end}} = \mathcal{I}^{t-1} \cup \{z_t, \hat{s}_t\}.$$

Their posterior belief about the persistent component is then

$$x_{t|t} \equiv \mathbb{E}[x_t | \mathcal{I}_t^{\text{end}}].$$

Agents update from the prior to the posterior using a Kalman filter:

$$(18) \quad x_{t|t} = x_{t|t-1} + \mathbf{k}'_{x,t}(\mathbf{s}_t - \mathbf{s}_{t|t-1}),$$

where $\mathbf{s}_t = [z_t, \hat{s}_t]$, $\mathbf{s}_{t|t-1} \equiv \mathbb{E}[\mathbf{s}_t | \mathcal{I}^{t-1}]$, and $\mathbf{k}'_{x,t}$ is the first row of the Kalman gain matrix. The details of the filtering problem are reported in Online Appendix Section B.1. The posterior formed at the end of period t determines the prior for the next period,

$$x_{t+1|t} = \rho x_{t|t}.$$

Key timing implication. The model therefore has a one-period information lag in decision-making. Period- t quantities are chosen using the prior $x_{t|t-1}$ and the history summarized by $\mathcal{I}^{t-1}$, while the current realizations $(z_t, \hat{s}_t)$ are observed only after those choices are fixed. As a result, the updated belief $x_{t|t}$ affects decisions only from period $t+1$ onward.

Assumption 1 *Firms and workers may observe publicly available real variables at time t (such as y_t, c_t, u_t, v_t , and w_t), but, apart from aggregate productivity z_t and the public signal $\hat{s}_t$, they do not use other endogenous variables to infer the aggregate state because processing those additional signals is costly. All learning about the persistent productivity component x_t is therefore summarized by the history of the pair $(z_\tau, \hat{s}_\tau)_{\tau \leq t-1}$ that enters the Kalman filter. In this sense, agents choose period- t controls using information available through the end of period $t-1$, and only update beliefs with $(z_t, \hat{s}_t)$ at the end of period t .⁹*

Optimization and equilibrium. Firms produce a homogeneous good with Cobb–Douglas technology $y_t = Z_t k_t^\zeta l_t^{1-\zeta}$, rent capital period by period, and choose capital and hiring intensity χ_t to maximize discounted profits net of wages, rental costs, and convex recruiting costs. Households pool income within families, choose consumption and savings, and workers choose search intensity when unemployed or when employed in bad matches. Unemployed workers accept both good and bad matches, while workers in bad matches search on the job and move only to good matches. Wages are determined by staggered

⁹This timing can be interpreted as aggregate data releases arriving with a one-period lag: data observed in period t summarize conditions in period $t-1$. In the model, belief errors are therefore analogous to nowcast errors.

Nash bargaining as in Gertler and Trigari (2009): with probability $1 - \lambda$ a contract is renegotiated, and otherwise the previous contract is retained.

The aggregate state at the beginning of period t is summarized by predetermined stocks and beliefs, $\{\bar{g}_t, \bar{b}_t, \bar{k}_t, \bar{w}_{t-1}, x_{t|t-1}, a_{t|t-1}\}$, together with the exogenous laws of motion in (14)–(17). A recursive equilibrium consists of value functions, prices, allocations, wage rules, and belief-updating rules such that households, workers, and firms optimize; search, matching, and wage contracts are consistent with individual decisions; markets clear; and beliefs evolve according to the Kalman filter with the timing in Assumption 1. Online Appendix Section B.6 lists the full Bellman equations, first-order conditions, wage-contracting equations, resource constraint, and recursive-equilibrium definition.

Special Cases

1. **Full-Information Benchmark.** The goal of the theoretical framework is twofold. First, to assess whether introducing imperfect information improves the prediction of the duration of the recovery of unemployment as compared to a full-information framework. Second, to understand the propagation mechanism for imperfect information. For either scenario, it is important to define the full-information benchmark. Under full information, the agents perfectly observe Z_t and x_t each period along with other variables. Therefore, when making their decisions, the agents are fully aware of the state of the economy and can perfectly observe each component of aggregate productivity. Hence there are only two shocks in this case: a persistent and a transitory productivity shock. With the same pre-commitment timing as the benchmark model, observed realizations at the end of period t are incorporated without signal-extraction error and determine expectations for period $t + 1$ decisions.
2. **Imperfect Information Without Noise.** Another important consideration is the role of imperfect information, even without noise shocks. In this framework, I assume that the information structure is the same as in the imperfect information with noise shocks framework, and agents observe z_t and a signal $\hat{s}_t$ about the persistent component of the aggregate productivity x_t . Here, the noise shocks are never realized, but agents believe that there is some noise in the economy and adjust their expectations accordingly.

Solution Method. To solve the model, I follow the approach in Gertler, Huckfeldt, and Trigari (2020). I first compute the deterministic steady state of the economy under imperfect information and then log-linearize the equilibrium conditions around this

steady state, including the firms' optimality conditions, the household Euler equation, the search and matching block, the wage-contracting equations, and the laws of motion for the composition of matches. The resulting linear rational expectations system can be written in the form

$$\Gamma_0 y_t = \Gamma_1 y_{t-1} + \Gamma_\varepsilon \varepsilon_t,$$

where y_t collects the relevant endogenous variables, ε_t is the vector of structural shocks, and Γ_ε maps those shocks into the linearized system. I solve this system using the Anderson–Moore (AIM) algorithm, which yields the unique stable solution for the policy functions. The imperfect-information structure is incorporated by augmenting the state vector with the agents' filtered beliefs about the persistent component of productivity and applying a standard Kalman filter to the linear state-space representation, given the signal process for $\hat{s}_t$. This yields decision rules expressed in terms of the perceived state, which I use to simulate the model, construct impulse responses, and compute the moments used in the calibration and estimation exercises.

4 Parameterization and Estimation

I estimate the parameters in the model at a quarterly frequency using a three-step procedure. First, I fix the parameters $\{\beta, \delta, \zeta, \omega, \lambda, \alpha, \eta_h, \rho\}$ to widely accepted values from the literature. Then, I estimate $\{\Psi, \kappa, \mu, \sigma, \phi, u^b, \xi\}$ by targeting unconditional stationary moments using the simulated method of moments (Table 4). Finally, the remaining information-process parameters $\{\sigma_\varepsilon, \sigma_v, \rho_a\}$ are estimated to match the impulse responses in Table 5.

Table 3 summarizes the externally fixed parameters.¹⁰ I calibrate the capital share $\zeta = 0.33$, the discount factor $\beta = 0.99$, and the depreciation rate $\delta = 0.025$ to widely accepted values in the business-cycle literature. The hiring cost convexity parameter is set to $\eta_h = 2.40$ following Merz and Yashiv (2007), which implies a degree of curvature in hiring costs consistent with their estimates.

Targeting Unconditional Moments.

As a first step, I target the steady-state unemployment rate, unemployment-to-employment

¹⁰I set the search-cost curvature parameter to $\omega = 3.60$ following Faberman et al. (2022). In the model, a higher ω makes the search-cost function $c(s) = \mu s^{1+1/\omega}$ less convex, implying a more elastic search-intensity margin. I therefore keep ω externally fixed at this micro-disciplined value rather than using it to absorb persistence in aggregate labor-market dynamics. I set workers' bargaining power to $\gamma = 0.50$ following Shimer (2005), set the non-renegotiation probability to $\lambda = 0.75$ following Gertler, Huckfeldt, and Trigari (2020), and set the matching elasticity with respect to searchers to $\alpha = 0.40$ following Gertler, Huckfeldt, and Trigari (2020). These are held fixed to preserve standard micro-disciplined values and avoid loading persistence-fit on bargaining, contract duration, or matching elasticity.

Table 3: Externally Fixed Parameter Values

Parameter	Interpretation	Value	Source
β	Discount factor	0.99	Shimer, 2005
δ	Depreciation rate	0.025	Gertler and Trigari, 2009
ζ	Capital share in production	0.33	Gertler, Huckfeldt, and Trigari, 2020
ω	Search-cost curvature parameter	3.60	Faberman et al., 2022
γ	Worker's bargaining power	0.50	Shimer, 2005
λ	Probability wage contract is not renegotiated	0.75	Gertler, Huckfeldt, and Trigari, 2020
α	Elasticity of matches w.r.t. searchers	0.40	Gertler, Huckfeldt, and Trigari, 2020
η_h	Hiring cost convexity	2.40	Merz and Yashiv, 2007
ρ	AR(1) coefficient of permanent productivity x_t	0.949	Gertler and Trigari, 2009

Note: This table reports externally calibrated parameter values drawn from the literature.

transitions, job-to-job transitions and the separation rate in the model to match the average values from the United States for the period 1968–2019 (Table 4). I also target the flow value of unemployment, u^b , to match the relative value of non-work to work activity $\bar{u}_T = 0.71$ following Hall and Milgrom (2008).¹¹

The efficiency parameter Ψ is chosen such that the steady-state unemployment rate in the model matches the average unemployment rate in the data for 1968–2019 and takes a value of 0.49. The hiring cost parameter κ determines the resources that firms invest in recruiting, and hence influences the probability that a worker finds a job. I set the steady-state job-finding probability to match the quarterly UE transition probability, $\tilde{p} = 0.28$, and then calibrate κ to be consistent with $\tilde{p}$. Furthermore, a higher search cost implies a lower EE probability and hence the search cost parameter μ is chosen to match the EE probability; it takes the value $\mu = 0.082$.

The job-survival probability σ is chosen so that the implied separation probability $1 - \sigma$ matches the $E-U$ transition probability. The steady-state productivity from a bad job, ϕ , is targeted to match the change in wages of workers who make job-to-job transitions. The ratio of bad jobs to good jobs is held constant and is calibrated following Gertler, Huckfeldt, and Trigari (2020).¹² I further calibrate ξ to match the average share of job transitions involving positive wage changes out of total job flows, targeting a value of 0.527 following Gertler, Huckfeldt, and Trigari (2020). The corresponding value is $\xi = 0.24$. A

¹¹Following Hall and Milgrom (2008), I calibrate u^b so that the model-implied steady-state value of non-work relative to work is $\bar{u}_T = 0.71$.

¹²Specifically, following Gertler, Huckfeldt, and Trigari (2020), the steady-state ratio of bad to good jobs satisfies

$$\frac{\bar{b}}{\bar{g}} = \frac{(1 - \lambda^{\text{surv}})(p^{EE} + p^{EU})}{p^{EE} + \lambda^{\text{surv}}p^{EU}},$$

where p^{EE} and p^{EU} are the probabilities of $E-E$ and $E-U$ transitions, respectively, and λ^{surv} is the job-survival parameter in their framework. In my calibration, I choose ξ so that the implied ratio $\bar{b}/\bar{g}$ matches this target given the model-implied transition probabilities.

Table 4: Parameters Estimated by Simulated Method of Moments

Parameters	Interpretation	Value	Target
Ψ	Match efficiency	0.49	Unemployment Rate = 0.055
κ	Cost of hiring	7.21	$U - E = 0.28$
μ	Scale parameter of search cost	0.082	$E - E = 0.025$
$1 - \sigma$	Separation rate	0.010	$E - U = 0.010$
ϕ	SS productivity from bad job	0.76	Average E-E wage increase = 0.045
ξ	Probability of finding a good job	0.24	Average wage-improv. flow share = 0.53
u^b	Flow value of unemployment	2.43	Relative value, non work = 0.71

Note: This table reports SMM estimates targeting key stationary labor-market moments: unemployment, unemployment-to-employment transitions, job-to-job transitions, employment-to-unemployment transitions, and wage changes for job-to-job movers. Moments are computed from CPS data over 1968Q4–2019Q4.

lower probability of finding a good job corresponds to a higher steady-state value of the bad-to-good worker ratio and hence a higher average share of bad-to-good flows. The resulting set of parameters estimated by SMM is reported in Table 4.

Information Parameters: Impulse Response Matching. I estimate the information parameters by matching model-implied responses following a noise shock and a TFP shock to their counterparts in the empirical exercise (Christiano, Eichenbaum, and Evans, 2005; Rotemberg and Woodford, 1997). The set of IRF targets is listed in Table 5 and includes the responses of the unemployment rate, the UE rate, and the EE rate for horizons of up to 20 quarters. The impulse response matching is done by minimizing the distance between the model-generated impulse response functions (IRFs) and the empirical IRFs. Let f be the column vector stacking the point estimates of each of these impulse responses across variables and horizons, and let $f_m(\vartheta)$ denote the corresponding model-generated IRFs, where ϑ is a vector of model parameters. The optimization problem is

$$(19) \quad \min_{\vartheta} (f - f_m(\vartheta))' W (f - f_m(\vartheta)),$$

where $\vartheta = \{\sigma_\epsilon, \sigma_v, \rho_a\}$. The weight matrix W is the inverse of the variance–covariance matrix of the empirical IRF estimates.¹³

The result of this estimation process is documented in Table 5, with standard errors calculated using the delta method (Guerron-Quintana, Inoue, and Kilian, 2017). These estimates imply a signal-to-noise ratio $\mathcal{K} = 0.23$, indicating that only a modest share of the variation in the signal reflects true innovations to the persistent component. This magnitude is in line with the noise-shock literature: Blanchard, L’Huillier, and Lorenzoni

¹³This objective is a GMM criterion with IRFs as moments and the inverse empirical IRF covariance matrix as the weighting matrix.

Table 5: Estimated Parameters from IRF Matching

Parameter	Interpretation	Estimate	Std. error
σ_ϵ	Std. dev. of persistent TFP shock	0.062	0.009
σ_ν	Std. dev. of noise shock	0.096	0.007
ρ_a	Persistence of noise component a_t	0.921	0.004
$\mathcal{K}$	Signal-to-noise ratio	0.23	0.073

Note: This table reports parameter estimates from the IRF-matching problem in equation 19. Column 3 reports point estimates and column 4 reports delta-method standard errors.

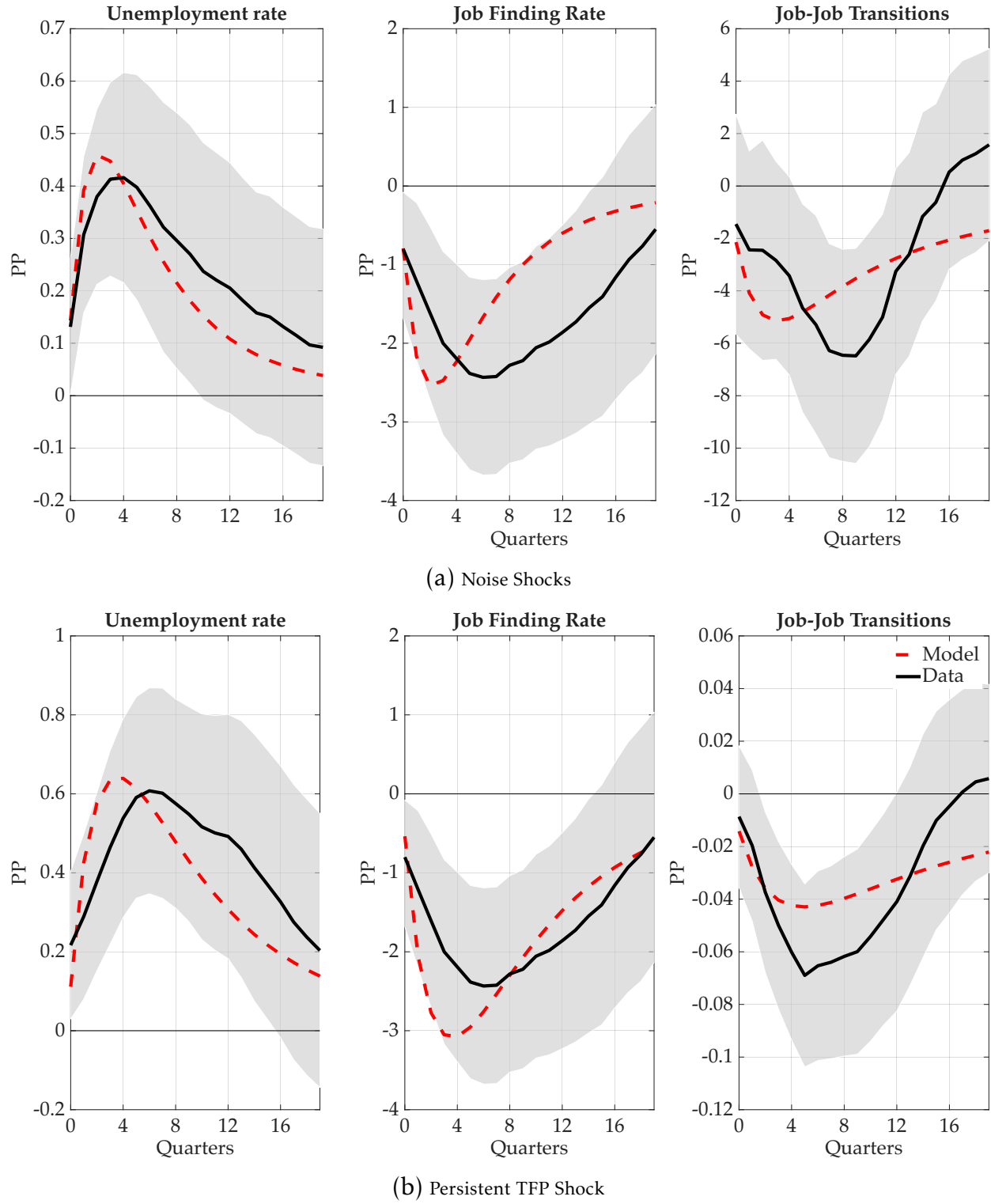
(2013) estimate a signal-extraction model in which noise accounts for a substantial share of short-run belief variation, and Coibion and Gorodnichenko (2015) imply a Kalman-gain weight on new information of similar order from SPF forecast errors. Within labor-market applications, Morales-Jiménez (2022) and Faccini and Melosi (2022) likewise find that a large share of signal variation is attributable to noise rather than fundamentals. The estimated persistence of the signal itself is high, $\rho_a = 0.92$, which slows learning considerably and amplifies the effects of noise shocks on labor-market dynamics. The model-implied impulse responses are plotted in Figure 5. Given externally fixed ρ and estimated σ_ϵ , I pin down σ_η from the unconditional variance of observed TFP using $\text{Var}(z_t) = \sigma_\epsilon^2/(1 - \rho^2) + \sigma_\eta^2$, which implies $\sigma_\eta = 0.192$. This keeps the model-consistent TFP volatility aligned with the data while the IRF matching identifies $\{\sigma_\epsilon, \sigma_\nu, \rho_a\}$.

Figure 5 plots the impulse responses from the empirical exercise and the impulse responses implied by the estimated model in response to a noise shock and a persistent TFP shock. The model fit is good, with all the model-implied impulse responses falling within the confidence bands from the empirical exercise. The early-horizon responses and overall dynamics for the unemployment rate, the job-finding rate, and job-to-job transitions match the empirical impulse responses well.

The benchmark model for the rest of the paper is the imperfect-information model with noise shocks. I consider several counterfactual models. The first is the full-information model, which is re-calibrated as discussed in Online Appendix Section B.2. In this framework, firms and workers perfectly observe z_t and x_t , so there is no signal-extraction error: realizations observed at the end of period t are used to form expectations for period $t + 1$ decisions. The second framework is the imperfect-information model without noise shocks. This model is not re-calibrated. Here, firms and workers still have imperfect information and do not observe x_t , but only see the signal $\hat{s}_t$ each period. However, the noise shocks are never realized. This model serves as an important comparison to highlight the role of imperfect information.

Mapping shocks from SVAR to the model. The SVAR noise shock is a reduced-form

Figure 5: Impulse Responses from Data and Estimated Model



Note: This figure compares model-implied IRFs (red dashed) with empirical SVAR IRFs (black solid) for noise and persistent TFP shocks. Shaded bands denote 95% empirical confidence intervals.

innovation to near-term beliefs, identified from GDP nowcast errors. In the model, the primitive disturbance is an innovation to the public signal about a latent persistent productivity component. These are not identical objects, but the model implies a disciplined mapping: signal shocks move nowcast errors when the signal is observed and, because decisions are based on prior beliefs, generate labor-market propagation with a short delay that mirrors the SVAR. Online Appendix Section B.3 validates this mapping by showing that the SVAR estimated on model-simulated data recovers the model-implied impulse responses.

Business Cycle Statistics To understand how the model with imperfect information performs with respect to the observed business cycle statistics in the data, I report the volatility and correlation of several labor market outcomes with output in Table 6. The table compares the business cycle statistics obtained by simulating the benchmark model and the full-information model to the statistics in the U.S. economy from 1968–2019 for the unemployment rate (U), job vacancies (V), job-to-job transitions (EE), job transitions from unemployment to employment (UE), and the hiring rate.

While both models match output dynamics reasonably well, the imperfect-information model outperforms the full-information model for all other variables, both in standard deviations and in correlations with output. It is worth noting that the full-information

Table 6: Business Cycle Statistics

x	Data		Full Information		Imperfect Info. without Noise		Imperfect Info. with Noise	
	(1) SD	(2) corr(Y,x)	(3) SD	(4) corr(Y,x)	(5) SD	(6) corr(Y,x)	(7) SD	(8) corr(Y,x)
Y	0.019	1	0.019	1	0.021	1	0.024	1
U	0.162	-0.859	0.121	-0.742	0.132	-0.768	0.151	-0.792
V	0.182	0.702	0.131	0.642	0.157	0.675	0.196	0.728
EE	0.102	0.720	0.067	0.629	0.071	0.661	0.088	0.825
UE	0.069	0.734	0.044	0.639	0.058	0.653	0.077	0.692
Hiring Rate	0.058	0.677	0.034	0.571	0.036	0.622	0.042	0.723

Note: This table reports model-implied standard deviations of key labor-market variables and their correlations with output. Simulated series are HP-filtered with smoothing parameter $\lambda_{HP} = 1,600$.

model already incorporates wage rigidity and on-the-job search, two forces known to amplify volatility in search models (Shimer, 2005).¹⁴ Yet introducing imperfect information raises unemployment volatility by an additional 25% relative to this benchmark. The imperfect-information framework likewise implies higher volatility for vacancies and transition rates, bringing the model closer to the data.

¹⁴In Table 6, the recalibrated full-information model generates unemployment volatility equal to about 75% of the data, consistent with Gertler, Huckfeldt, and Trigari, 2020.

Forecast Error Variance Decomposition The identified noise shocks explain about a third of the variance in labor market variables at short-run horizons. To understand how the benchmark model compares to the observed moments, I report the forecast error variance decomposition calculated by simulating the imperfect-information model, in Table 7, for an 8-quarter horizon. The benchmark model matches the forecast error variances of the key labor market outcomes observed in the data reasonably well. The model predicts that noise shocks explain 31% of the forecast error variance in the unemployment rate, which is about 90% of the corresponding share in the data. On average, the model overpredicts the forecast error variance by 7%.

Table 7: FEVD: Data and Model

	Data			Model		
	Horizon: 0-8 quarters			Horizon: 0-8 quarters		
	Persistent	Transitory	Noise	Persistent	Transitory	Noise
Unemployment	0.43	0.23	0.34	0.48	0.21	0.31
Vacancies	0.42	0.21	0.37	0.49	0.19	0.32
<i>UE</i>	0.38	0.27	0.35	0.42	0.21	0.37
<i>EE</i>	0.42	0.31	0.27	0.49	0.17	0.34
Wage Growth	0.61	0.25	0.14	0.60	0.22	0.18

Note: This table reports forecast error variance shares from the imperfect-information model and compares them with their data counterparts.

5 Role of Imperfect Information in Labor Market Dynamics

In this section I document that the calibrated model successfully matches the observed persistence of the unemployment rate. Specifically, it takes 14 quarters in the model for unemployment to reverse 75% of its recessionary increase, while the corresponding number in the data is 17 quarters.¹⁵ In the model without imperfect information it would take only 9 quarters, so imperfect information adds about 5 quarters. This additional persistence is generated through two channels. First, learning under imperfect information generates endogenous persistence: agents under-react to productivity shocks when forming the next period's decisions. Together with sticky wages, this causes a delay in labor-market adjustment and thus increases persistence. Second, noise shocks themselves may prolong recoveries, because once they are incorporated into beliefs, agents can mistake them for

¹⁵This 17-quarter target is the average 75%-recovery horizon across the 1973–75, 1981–82, 1990–91, 2001, and 2007–09 recessions (Online Appendix Table A2; the short-lived 1980 recession is excluded). I use the 75% threshold because the 2001 recession does not return to its pre-recession trough before the next downturn.

actual productivity shocks and adjust subsequent behavior accordingly, even though fundamentals are unchanged.

After establishing the mechanism, I present a counterfactual exercise that demonstrates how imperfect information can contribute significantly to the slow recovery of the unemployment rate during recessions, and show that the model with imperfect information can closely match the unemployment path across several postwar recoveries. Finally, I discuss the importance of other channels proposed to generate persistence in search-and-matching models, such as sticky wages and on-the-job search. I find that while sticky wages alone can generate about 45% of the observed persistence, its interaction with imperfect information accounts for about 65% of the persistence, highlighting the importance of that interaction.

5.1 Mechanism for Propagation of Shocks under Imperfect Information

Figure 6 illustrates the effect of a one-standard-deviation negative persistent productivity shock on key outcomes. The solid lines represent the imperfect-information framework, while the dashed lines are the full-information benchmark where firms and workers can perfectly observe the persistent and transitory components of aggregate productivity each period. In an environment characterized by imperfect information, agents—both firms and workers—operate under Bayesian learning: they assign probabilistic weight to shocks being persistent, transitory, or merely noise. With the timing described in Section 3, current shocks are observed only at the end of the period, so the key difference between full and imperfect information appears in the subsequent path of hiring, search, and wage-setting rather than in a same-period jump in controls. I find that incorporating imperfect information about shock persistence increases the persistence of unemployment by about 30% relative to the full-information model; this operates through two channels.

Channel 1. The response to a persistent productivity shock becomes more persistent under imperfect information. Consider a negative persistent shock. Once period- t outcomes are observed, full-information agents know that the shock is persistent, whereas imperfect-information agents still attach positive probability to it being transitory or noise. This makes the response of period- $t+1$ hiring, search, and wage-setting more muted than under full information, an under-reaction that is amplified by wage rigidity. Sticky wages make agents reluctant to adjust wages quickly because if the shock turns out to be transitory or noise, such adjustments would be suboptimal. Relative to a full-information model, wages, vacancies, and search effort therefore decline more gradually, producing an overall delay in the labor-market response.

As agents update beliefs and place more weight on the shock being persistent, wages negotiated at renegotiation opportunities fall and average wages decline. Unemployed workers reduce search effort as expected returns from employment decline; employed workers reduce their upward search intensity; and firms post fewer vacancies as they expect lower returns from hiring. These forces lower matching efficiency and job-finding rates, generating a hump-shaped increase in unemployment that is more persistent than under the full-information benchmark.

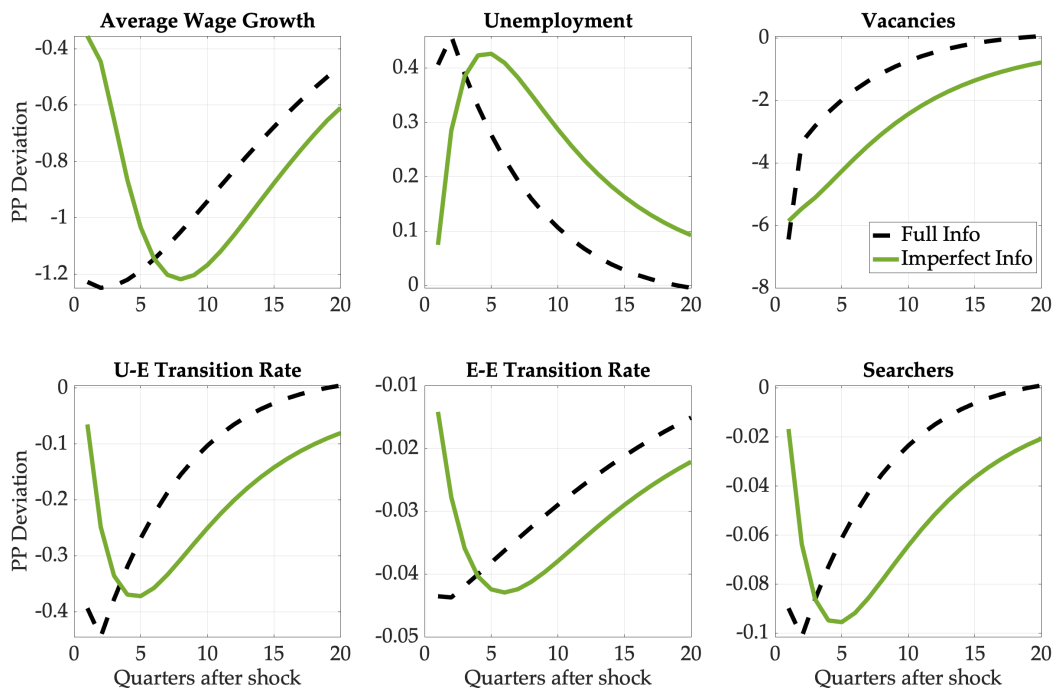
This channel crucially depends on sticky wages: with fully flexible wages, adjustments would be quicker as agents learn, and recoveries would be faster. On-the-job search further amplifies this effect because employed workers occupy low-quality matches and crowd out unemployed workers; firms then post fewer vacancies, further delaying unemployed workers' job-finding.

Channel 2. Noise shocks provide an independent source of persistence. A negative noise shock does not change fundamentals and, under the model timing, does not alter period- t controls on impact. However, once the signal is observed at the end of the period, agents partially attribute the perceived deterioration to an actual change in persistent productivity. In the next period firms expecting lower returns post fewer vacancies; unemployed workers reduce search intensity anticipating lower wages; and employed workers cut upward search. Together these belief-driven responses reduce matches and raise unemployment despite the absence of a fundamentals decline.

Both channels interact and are amplified by sticky wages and on-the-job search. Because choices respond with a one-period lag, what matters quantitatively is the sequence of belief revisions generated by persistent, transitory, and noise shocks. When agents receive such a sequence, they tend to overestimate the persistence of negative shocks in the presence of noise and behave as if facing a more persistent decline than actually occurred. Firms therefore keep vacancies lower for longer, and the resulting congestion and lower search intensity slow the recovery of the unemployment rate.

During downturns, both channels can act together and amplify unemployment persistence. Agents tend to overestimate the persistence of negative productivity shocks in the presence of noise shocks and therefore behave as if the decline is more persistent than it actually is. As they slowly learn whether shocks are persistent or transitory, their period- $t+1$ and subsequent decisions reflect an outlook that is too pessimistic relative to fundamentals. Firms therefore expect lower future revenue and post fewer vacancies for longer than warranted by fundamentals. The decline in vacancies reduces job-offer arrival rates for both the unemployed and the employed, who lower their search effort as

Figure 6: Impulse Response to a Negative Persistent TFP Shock



Note: This figure shows the impulse response functions for the recalibrated full-information model (dashed black line) and the imperfect-information model (solid green line) to a negative one-standard-deviation persistent TFP shock.

the returns to search fall. Because many bad matches remain occupied and good jobs are scarce, congestion at the lower rungs of the ladder further depresses unemployed workers' job-finding rates, keeping unemployment elevated longer than implied by fundamentals.

It is important to note that the imperfect-information mechanism is not intrinsically asymmetric. By slowing the adjustment of beliefs to both negative and positive productivity innovations, the mechanism would also tend to prolong booms, not just recessions. Thus, the asymmetry emphasized in the data should be understood as coming from the realized mix of shocks—in particular, the sequence and relative importance of persistent and noise shocks during downturns—rather than from an asymmetric propagation mechanism in the model itself. My focus on recession episodes reflects the empirical motivation of the paper and the policy relevance of slow unemployment recoveries, rather than any asymmetry built into the model.

5.2 Unemployment Dynamics across Recessions: Data vs Model

In this section I simulate the calibrated imperfect-information model to generate counterfactual unemployment series for five recessions between 1968 and 2019. This exercise

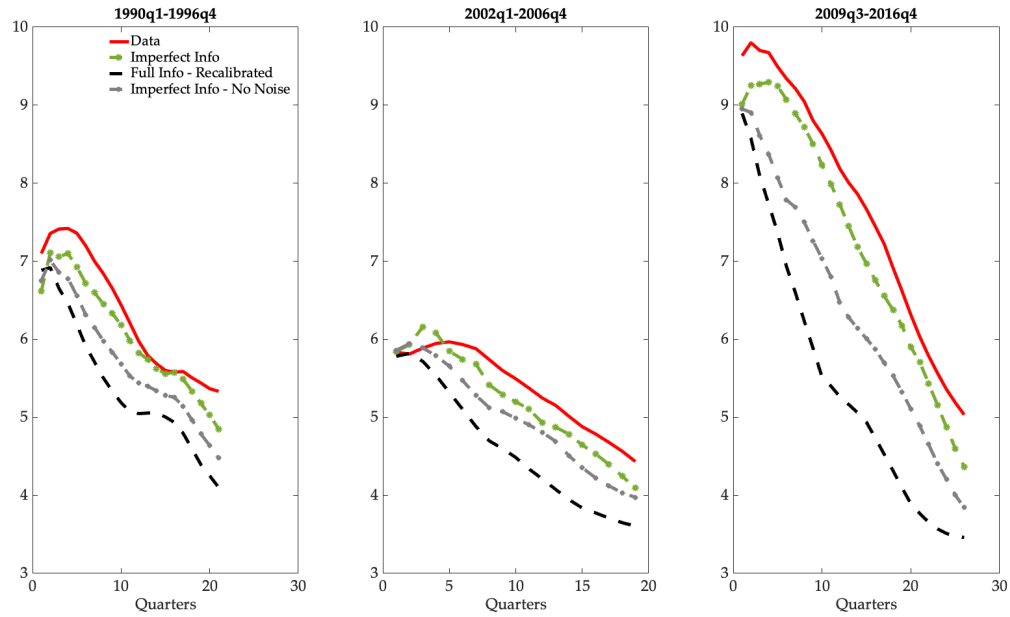
shows that imperfect information helps explain the slow recovery of unemployment, especially in the last three recessions. The VAR identification delivers three structural innovations: a persistent TFP shock, a transitory TFP shock, and a noise (belief) shock that moves expectations without contemporaneously affecting TFP. In the model, these map directly into the productivity and signal processes in equations (14)–(17). The persistent TFP innovation corresponds to ϵ_t in the law of motion for the persistent component x_t in (15); the transitory TFP innovation corresponds to η_t in (14); and the identified noise shock corresponds to v_t in the noise term a_t in (17), which enters the public signal $\hat{s}_t = x_t + a_t$ in (16). Given the pre-commitment timing, these shocks are observed at the end of period t and influence labor-market choices from period $t+1$ onward. Thus, when I feed persistent, transitory, or noise shocks into the model, their timing and relative variances are disciplined by the same objects identified in the data. For each counterfactual exercise, I normalize the model to match the starting unemployment rate. When simulating the imperfect-information model with noise, all three VAR shocks are fed into the model each period; the imperfect-information model without noise uses only the persistent and transitory shocks. For the full-information model I introduce the persistent and transitory shocks each period and recalibrate the model to match the empirical IRFs to persistent shocks. Estimated parameters for the full-information model appear in Online Appendix Section B.2.

The imperfect-information model captures the persistence of unemployment well for the three recessions after 1990. In contrast, the full-information model predicts much faster recoveries. For the Great Recession, the unemployment rate took 33 quarters to return to its pre-recession trough; the imperfect-information model predicts 32 quarters, while the full-information model predicts 22 quarters (about 10 quarters faster than the imperfect-information prediction).

For the 2001 recession, the unemployment rate does not return to its pre-recession trough before the next recession (NA in the data); the imperfect-information model predicts 21 quarters and the full-information model 14 quarters. In the 1990–91 recession, the data show 21 quarters; the imperfect-information model predicts 24 quarters while the full-information model predicts 16 quarters. These differences underscore the contribution of imperfect information: larger noise shocks make agents perceive negative productivity shocks as more persistent than they are, and slow learning combined with sticky wages produces slower recoveries.

For pre-1990 recessions the full-information and the two imperfect-information models give similar predictions because the noise shocks identified in the SVAR play a smaller role. With smaller noise shocks, agents’ perceived productivity is closer to

Figure 7: Model-Implied Recovery of Unemployment for Recessions Post 1990s



Note: This figure shows model-implied simulated unemployment for the recalibrated full-information model (dashed blue), the imperfect-information model without noise (gray), and the imperfect-information model with noise (solid green) for the Great Recession, the 2001 recession, and the 1990–91 recession.

true productivity and recoveries are similar across models. For example, in the 1981–82 recession (22 quarters), the imperfect-information model with noise predicts recovery in 21 quarters while the full-information model predicts 17 quarters. In the 1973–75 recession, the data do not reach full recovery before the next downturn (NA), while the imperfect-information model predicts 17 quarters and the full-information model predicts 14 quarters. Results are summarized in Online Appendix Table B2, and Online Appendix Figure B6 reports the full set of recession simulations.

The quantitative importance of noise shocks is especially pronounced in post-1990 recessions, suggesting that imperfect information has become more relevant in recent decades. At the same time, different downturns may differ in the mix of fundamental and belief-driven forces. In the current analysis, the shock processes are held fixed across episodes; allowing information precision or the relative variance of shocks to vary across recessions would help connect the model more tightly to the historical heterogeneity highlighted in the empirical decompositions.

5.2.1 Model Decomposition: Sticky Wages, On-the-job Search and Imperfect Information

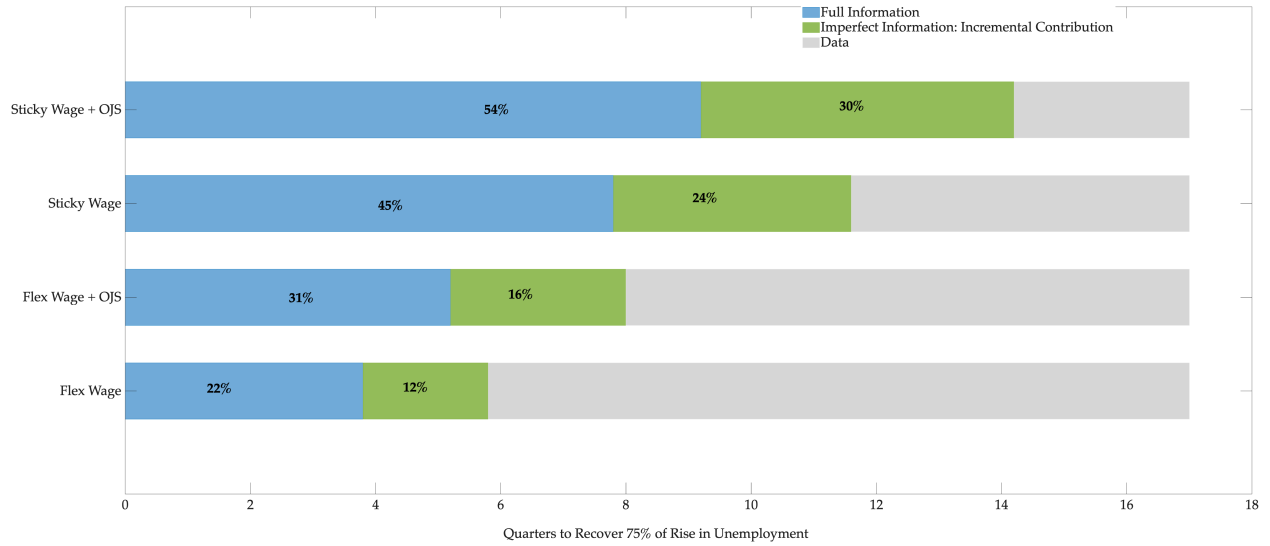
In this section I compare unemployment persistence across mechanisms with and without imperfect information. The model contains three sources of persistence: on-the-job search, sticky wages, and learning. To quantify each channel, I compare a full-information benchmark to the imperfect-information model under four scenarios: (a) flexible wages without on-the-job search, (b) flexible wages with on-the-job search, (c) sticky wages without on-the-job search, and (d) sticky wages with on-the-job search.

Unlike the recession-by-recession comparisons above, this decomposition uses a 75%-recovery metric. The persistence measure is the average number of quarters across recessions required for unemployment to reverse 75% of its recessionary increase. For each recession between 1968 and 2019 I compute the share of the recessionary rise in unemployment that is reversed during the subsequent expansion, and record the number of quarters until the 75% threshold is reached. Empirically this average is 17 quarters, computed across the 1973–75, 1981–82, 1990–91, 2001, and 2007–09 recessions (footnote 1 and Online Appendix Table A2; the short-lived 1980 recession is excluded). I use 75% rather than full recovery because the 2001 recession does not return to its pre-recession trough before the next downturn. To show how learning contributes under each specification I plot this measure in Figure 8. In the stacked bar graph I recalibrate the full-information model as described in Online Appendix Section B.2.

Role of sticky wages. Sticky wages generate persistence by making downward wage adjustment sluggish. When wages adjust slowly during recessions, firms’ incentives to hire remain low, hiring falls, and job-finding rates stay depressed longer than under flexible wages. Introducing sticky wages without on-the-job search accounts for about 45% (≈ 7 quarters) of the 75%-recovery duration under the full-information benchmark.

Role of on-the-job search. On-the-job search produces congestion: during downturns more employed workers remain in bad matches and unemployed workers have fewer low-productivity vacancies to move into. Employed workers also search less intensively, which leads firms to post fewer vacancies and further reduces unemployed workers’ job-finding probabilities. In a flexible-wage full-information model, introducing on-the-job search explains about 30% (≈ 5 quarters) of the recovery duration; with sticky wages the full-information model predicts about 54% (≈ 9 quarters).

Figure 8: Average Duration to Reverse 75% of the Recessionary Increase in Unemployment Across Models



Note: This figure shows the model-implied average duration (quarters) for unemployment to reverse 75% of its recessionary increase, averaged across recessions between 1968 and 2019 for various model specifications. Percentages indicate the share of the empirical duration (17 quarters) explained by each specification. The green segments show the incremental contribution of learning; the total imperfect-information contribution is the sum of the blue and green bars.

Role of imperfect information. The previous two paragraphs show that on-the-job search and sticky wages already produce considerable persistence—together they predict 75% recovery in about 9 quarters under the full-information benchmark. To quantify imperfect information’s added effect, consider that during downturns noise shocks make agents perceive productivity shocks as more persistent than they are once those signals are incorporated into beliefs. Because agents learn only gradually whether a shock is persistent or transitory, they behave as if facing a more persistent negative shock for several periods. Firms therefore lower hiring and workers reduce search effort in subsequent periods, reducing matches until learning occurs. Adding imperfect information to the flexible-wage, no-on-the-job-search case increases persistence, predicting 75% recovery in about 6 quarters ($\approx 35\%$ of the total).

Learning interacting with on-the-job search adds roughly another 16% to persistence, yielding 75% recovery in about 8 quarters. Employed workers learn slowly and continue to expect worse productivity, so their search effort remains dampened; firms post fewer vacancies, and unemployed workers’ job-finding probabilities remain lower for longer.

Learning interacting with sticky wages raises persistence even without on-the-job search. Firms anticipating persistently lower productivity reduce hiring, and slow wage adjustments further dampen hiring incentives. Workers cut search effort, and together these forces depress job-finding rates for longer. Imperfect information with sticky wages predicts an 11-quarter horizon to reverse 75% of the recessionary increase ($\approx 65\%$ of the

empirical duration), about 24% longer than the sticky-wage full-information model.

Finally, imperfect information combined with sticky wages and on-the-job search adds about 5 quarters ($\approx 30\%$) to the average 75%-recovery duration relative to the full-information counterpart. In equilibrium, agents' anticipation of persistently lower productivity depresses hiring and search; sticky wages and reduced on-the-job search amplify these effects, jointly generating roughly 84% of the observed persistence.

In Online Appendix Section B.5.4 I present a decomposition comparing the full-information model with the imperfect-information model without noise shocks. That analysis shows that learning alone increases unemployment persistence even without noise shocks: in the sticky-wage, on-the-job-search model imperfect information adds about 18% (≈ 3 quarters) to persistence. This underscores the importance of incorporating imperfect information in search-and-matching models to better predict recovery durations.

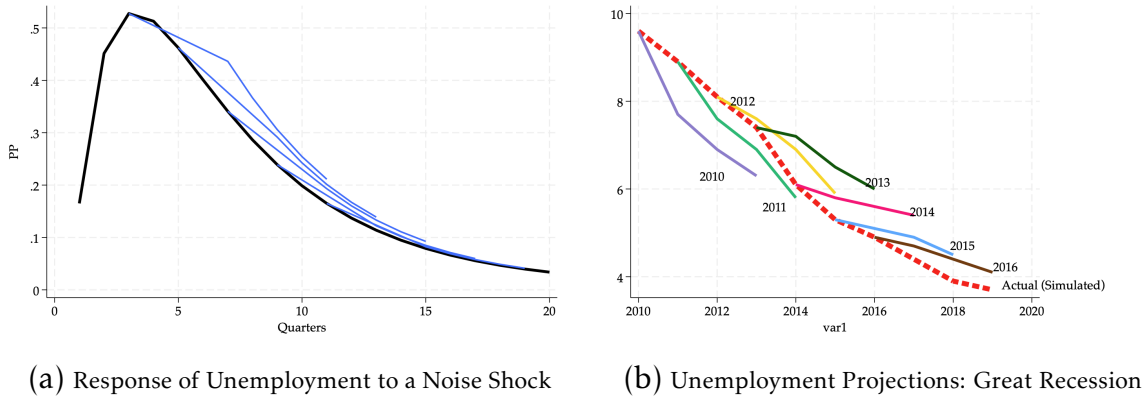
5.2.2 Unemployment Forecasts in the Model.

The model implies distinctive forecast dynamics. Under a persistent productivity shock agents under imperfect information attribute part of the shock to noise or the transitory component, causing their projections to under-react to the subsequent rise in unemployment. Conversely, under a noise shock agents may attribute some of it to a persistent or transitory productivity decline and initially forecast unemployment to be higher than it will be over the following quarters; as they learn that the shock is noise, forecasts converge to the realized unemployment path. Figure 9 illustrates this: Panel (a) shows 4–8-quarter-ahead forecasts in response to a noise shock together with the impulse response of unemployment; Panel (b) shows actual and forecasted unemployment after the Great Recession.

To show that noise shocks can reproduce the overshooting of unemployment projections observed in the data, I simulate long-run forecasts after the Great Recession. When the persistent shock contribution dominates, projections under-react; when the noise contribution dominates, projections overestimate unemployment. The historical decomposition in Figure A13 indicates that noise shocks' contribution to unemployment movements rises after 2012. In the model, this generates initial underestimation when persistent shocks dominate and later overestimation as noise becomes more important, mirroring the pattern in the data (Figure 2b). Note that noise shocks are unique in producing long-run overestimation; for other structural shocks long-run expectations tend to underestimate unemployment.

For additional evidence from the COVID-19 recession, including the SVAR extension and forecast behavior, see Online Appendix Section A.4.1.

Figure 9: Long-Run Unemployment Projections in the Model



Note: Panel (a) shows model-implied 4–8-quarter-ahead projections in response to a noise shock; the solid thick black line is the actual unemployment response. Panel (b) shows model-implied one-, two-, and three-year-ahead forecasts for the Great Recession. While simulating, all three VAR-identified shocks act each period.

6 Conclusion

This paper documents that noise shocks—expectational errors arising from imperfect information about productivity persistence—are an important driver of sluggish labor market recoveries. Using a tri-variate structural VAR, I show that absent noise shocks, unemployment would have recovered most of its recessionary rise six quarters earlier on average across postwar recessions. Noise shocks account for about one-third of unemployment variance at business cycle frequencies. I embed this information structure into a search-and-matching model with staggered wages, convex hiring costs, and on-the-job search. The calibrated model generates approximately 30 percent more persistence in unemployment recoveries above and beyond the full-information benchmark. The mechanism operates through two channels: after period- t outcomes are observed, slow learning causes agents to under-react in subsequent decisions to persistent shocks, while noise shocks lead them to overestimate the persistence of downturns. Sticky wages and on-the-job search amplify both channels. These effects are especially pronounced in post-1990 recessions.

Several directions for future work are promising. Extending the framework to include heterogeneous households and precautionary savings would allow imperfect information to interact with demand-driven amplification. Allowing for information asymmetries across firms and workers could generate additional cross-sectional dispersion in beliefs and further shape aggregate dynamics. Finally, using micro-level expectation data from forecasters, firms, and households to discipline the information structure more tightly is a natural next step.

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A Paper Appendix: Key Robustness Checks

Online Appendix: <https://anushkamitra.com/research/JMP-Mitra-2023/JMP-Mitra-2023-online-appendix.pdf>

This appendix reports the robustness checks and supporting results most directly related to the paper’s interpretation of noise shocks. The Online Appendix contains the full set of auxiliary results, model equations, and additional figures.

A.1 Labor-Market Flows During Recoveries

The slow post-1990 recoveries are visible not only in unemployment stocks, but also in worker-flow measures. Online Appendix Table A1 reports the recovery-horizon exercise for the job-finding and separation rates. Job-finding rates recover especially slowly after the 1990–91, 2001, and 2007–09 recessions, while separation rates tend to normalize more quickly. This pattern supports the paper’s emphasis on sluggish vacancy creation and hiring as central margins of slow labor-market recoveries.

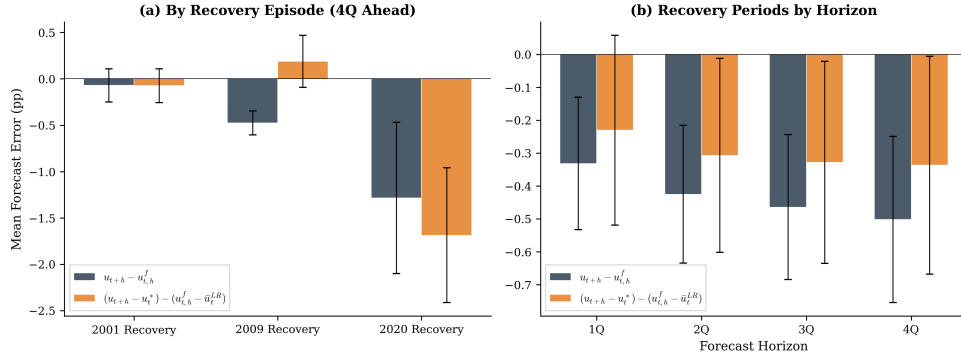
A.2 Natural-Rate Misperceptions

A possible alternative explanation for unemployment forecast errors is that forecasters misperceive the natural rate of unemployment rather than the cyclical persistence of shocks. Figure 10 compares standard SPF unemployment forecast errors with a gap-adjusted measure that removes disagreement between SPF long-run unemployment expectations and the CBO natural rate. The adjustment reduces mean forecast errors by about one-third on average across recoveries, but does not eliminate them, leaving a residual consistent with misperceived cyclical persistence.

A.3 Noise versus News

A second concern is that the identified noise shocks may instead capture news about future productivity. Figure 11 shows that the identified noise shock moves nowcast errors and

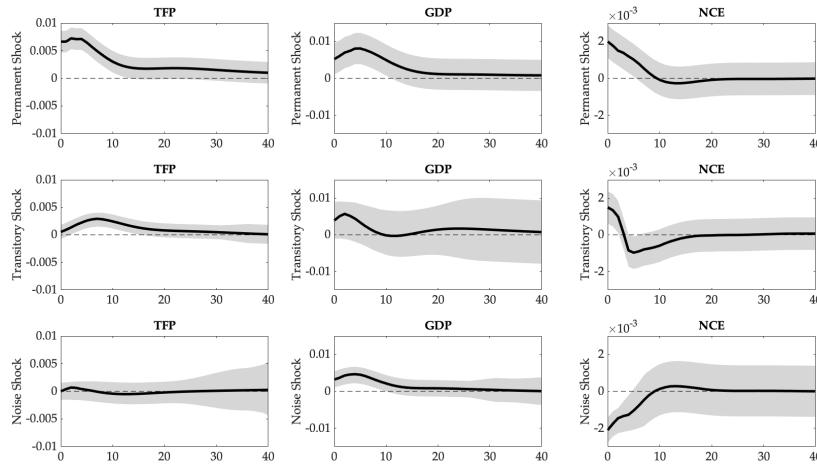
Figure 10: Unemployment Forecast Errors: Natural Rate Adjustment



Note: Dark bars report mean SPF unemployment forecast errors. Orange bars report forecast errors adjusted for differences between SPF long-run unemployment expectations and the CBO natural rate. Negative values indicate overprediction.

real activity but does not generate systematic movements in measured TFP. This pattern supports the interpretation of the disturbance as a non-fundamental information shock rather than advance information about future productivity.

Figure 11: Impulse Responses from the SVAR



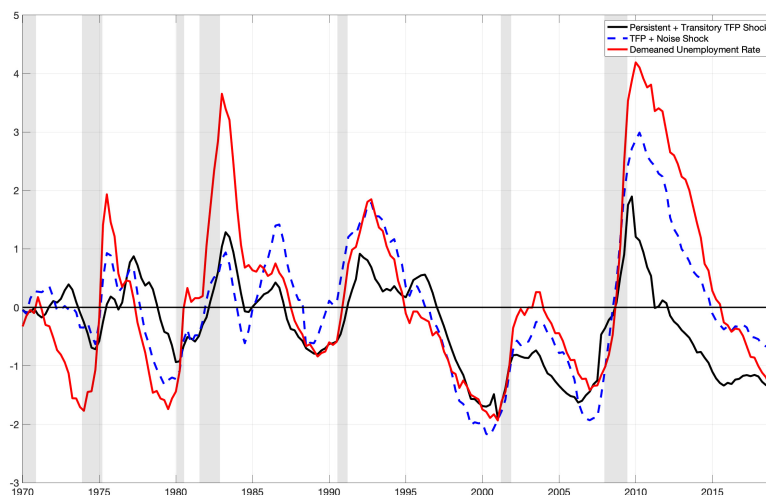
Note: Impulse responses to persistent, transitory, and noise shocks identified by the SVAR. Baseline sample: 1968Q4–2019Q4.

A.4 Allowing for an Unrestricted Shock

To assess whether omitted shocks drive the baseline decomposition, I estimate a four-variable VAR that adds unemployment and leaves a fourth shock unrestricted. Figure 12 shows that noise shocks still explain a substantial part of unemployment fluctuations in

the 1990–91, 2001, and 2007–09 recessions. The unrestricted shock is more important in the 1970s and early 1980s, consistent with a larger role for non-TFP disturbances in those episodes.

Figure 12: Four-Variable VAR: Historical Contribution to Unemployment



Note: The figure reports the historical decomposition of unemployment in a VAR with TFP, GDP growth, nowcast errors, and unemployment. The fourth shock is unrestricted.

A.5 Controlling for Uncertainty Shocks

Finally, I re-estimate the labor-market responses to noise shocks while controlling for uncertainty shocks from Bloom (2009). The uncertainty-controlled responses remain close to the baseline estimates and within the confidence bands. The persistence and shape of the noise-shock responses are therefore not driven by standard uncertainty-shock controls. The corresponding figure and full robustness discussion are reported in Online Appendix Section A.7.